Eduardo de la Rosa Flores

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Professional Objective

Mechatronics Engineer focused on **embedded firmware testing and validation**: writing test cases, building automated test benches, and verifying real-time control firmware. Hands-on with **Hardware-in-the-Loop (HIL) validation**, automotive serial protocols, and hardware debugging, backed by automotive data quality-management experience at BMW Group.

Education

Instituto Tecnológico y de Estudios Superiores de Monterrey — *Monterrey, MX*

B.S. in Mechatronics Engineering — Aug. 2020 – Jun. 2026

- **GPA: 90/100** | *DHIK Double Degree Program, scholarship awarded for academic excellence.*
- **Relevant Courses:** Firmware Testing, Communication Protocols, Industrial Automation, Robotic Software.
- **Extracurricular:** German Teacher & Study Abroad Mentor

Esslingen University of Applied Sciences — *Esslingen, DE*

B.Eng. in Automation Systems and Production Informatics — Sep. 2024 – Feb. 2026

- **Relevant Courses:** AI and Machine Learning, Project Management and Robotics.

Skills

Languages: Spanish (Native), English (C1), German (B2), French (A2)

Soft Skills: Systems Thinking – Assertive & Efficient Communication – Adaptability – Leadership – Continuous Learning

Technical Skills:

Test & Validation: HIL Testing, Test Cases & Test Plans, Test Automation, Hardware Debugging, Oscilloscopes & Logic Analyzers, Fault Diagnostics

Embedded Systems: Microcontroller Architectures, RTOS (FreeRTOS), Real-Time Control, Serial Protocols (CAN, I2C, SPI, UART)

Software & Tooling: Python, C/C++, MATLAB, Git, CI/CD, Docker, Agile/Scrum, PLC Programming

Experience

BMW Group — *Munich, DE*

Data Engineering & AI Integration Intern – *Thesis in Supply Chain Quality Management* — Mar. 2025 – Jan. 2026

- Built an AI pipeline (**GPT-4.1 & PySpark**) that automatically matches factory production failures to issues already seen in **early testing phases** at **94% accuracy**, replacing manual cross-checks with traceable root-cause links.
- Centralized fragmented supplier-quality data from **5 separate databases** into a single **Palantir Foundry** model, enabling end-to-end traceability of parts and suppliers across the full lifecycle, before and after start of production (SOP).
- Automated **Python ETL pipelines** and interactive dashboards over hundreds of thousands of quality records, replacing slow manual report reading with real-time insight, used by a **15-engineer team** and scaled across multiple business units.
- Built and deployed a cloud-based gamified web app integrating the **Google Fit REST API** for live team leaderboards.
- Completed a hands-on **assembly-line rotation on underbody components**, gaining first-hand insight into how physical manufacturing defects actually surface, which directly informed the quality-tracking logic I built upstream.

Éxodo Al-Melek — *Monterrey, MX*

Regional Subcoordinator — Jul. 2021 – Jul. 2024

- Led a multidisciplinary team of 20 people to **plan, organize, and execute regional events** for 150+ participants.
- Adapted communication styles across multi-generational groups to build trust and ensure alignment.

Technical Projects

HIL Test Bench for Power Management – *Industry Partner: International Motors Mexico*

- Designed a 3-layer HIL test bench with **galvanic isolation** and a solid-state switching matrix for safe automated power cycling of 4 independent ECU channels (12V/5A), shielding the 3.3V control logic from 12V transient spikes during continuous stress-testing.
- Developed event-driven **asyncio MicroPython firmware** on the ESP32 that parses SCPI/VISA commands while concurrently driving an SPI-CAN controller and polling bare-metal I2C drivers, streaming channel voltage, current, and power telemetry simultaneously.
- Built a multithreaded **Python/Tkinter** host application with serial-connection watchdogs, live oscilloscope plotting, and a CAN monitor for ECU node supervision, plus a scripting engine that parses LOOP/DELAY automation files into SCPI/VISA stress-test sequences.

Dual-Core Smart Tire Inflation Platform – *Industry Partner: John Deere – Advanced Embedded Systems*

- Engineered a dual-core **asymmetric multiprocessing (AMP) architecture on STM32H7** (Cortex-M7 @480MHz + Cortex-M4 @240MHz), using **hardware semaphores** for a deterministic boot handshake and conflict-free cross-core peripheral access.
- Ran a discrete positional **PID loop** on the CM7 core updating every 500ms, mapping a 16-bit ADC pressure transducer into compressor and solenoid PWM duty cycles, with safety interlocks that saturate the valve output when pressure exceeds 110% of the target setpoint.
- Owned the **FDCAN1** peripheral in Classic CAN at 500 Kbps with hardware masking filters to receive setpoints from the tractor bus, and ran an independent **FreeRTOS + FATFS** supervisor on the CM4 core logging operational blackbox parameters to an SD card.

Autonomous 4WD Mobile Robot – *Industry Partner: John Deere – Advanced Control*

- Performed **ARX plant identification** in MATLAB (>70% accuracy) and used the model to design a cascaded control architecture on an **STM32**: 4 independent speed PI loops (12-bit PWM, 4 EXTI encoder lines) coordinated under a master orientation PI loop.
- Drove the master orientation PI tracking loop with high-pass filtered **MPU6050 gyroscope** telemetry and suppressed encoder noise through a digital moving-average filter, stabilizing the deterministic closed-loop response of the 4WD skid-steer platform at 72MHz.

RTOS Automotive Infotainment System – *RTOS & Multithreading*

- Engineered a deterministic 4-task real-time firmware flow on an **STM32F103** using thread-safe message queues to eliminate UI blocking during I/O, in a distributed client-server architecture pairing FreeRTOS with a Raspberry Pi multimedia engine over UART.
- Drove a 4x4 matrix keypad and a 4-bit 16x2 LCD from the STM32 client node, while the **Raspberry Pi server** ran a Python **pygame.mixer** daemon mapping keystrokes to play, pause, skip, and volume controls over an .ogg library.